ROUGH DRAFT

Beneath the Mesa

Assessing Aquifer and Well Water Resources Across Sandoval County, New Mexico

A County-Scale Screening Study with Cuba, New Mexico as a Focus Area

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ABSTRACT

Water scarcity is a persistent challenge across New Mexico. Sandoval County spans several hydrogeologic settings, and its communities depend on a mix of groundwater wells, surface water, and acequia irrigation systems. This project focuses on groundwater: it evaluates the condition of the county's aquifers and wells, with Cuba, New Mexico, on the eastern edge of the San Juan Basin, as a focused case study. The project is motivated by concern about declining groundwater near Cuba, but rather than assume how severe that decline is or what is causing it, we test what the available public data actually show, and where they are too sparse to say. *[Add the data used, the main findings, and what they mean, in three to four sentences, once the analysis is complete.]*

1 Introduction

1.1 Background and Motivation

[Background on water resources in New Mexico and Sandoval County, and the concern about declining groundwater near Cuba that motivated the project.]

1.2 Approach

The study works at two levels:

- **County-wide screening.** Map and compare aquifers, wells, and groundwater levels across Sandoval County. Because the county crosses several geologic regions, comparisons are grouped by aquifer unit instead of treating the county as one uniform aquifer.
- **Cuba case study.** A closer look at the Ojo Alamo Sandstone and San Jose Formation near Cuba, using published maps of both formations, aquifer-test records from nearby wells, and stream measurements on the Rio Puerco at Cuba.

1.3 Research Questions

RQ1. How have groundwater levels changed at monitored wells in Sandoval County, and which wells show sustained decline, stability, or recovery?

RQ2. What do well records show about well depth, depth to water, and the aquifer units that wells in different parts of the county draw from?

RQ3. What do available records show about groundwater near Cuba, and are they enough to evaluate the concern that its main aquifer is approaching severe depletion?

RQ4. How do the Ojo Alamo Sandstone and San Jose Formation compare near Cuba in extent, thickness, and hydraulic properties?

RQ5. Where is monitoring coverage strong, and where are the most important data gaps?

RQ6. Which indicators can be combined into a dashboard that shows groundwater stress without overstating how certain the data are?

The project proposal also asked about groundwater quality, acequias, and links between groundwater and surface water. Those questions were dropped because the data needed to answer them were not available (Section 7).

Figure 1 (to be added by the analysis lead): county map showing water-level monitoring wells, aquifer-test wells colored by formation, and the Rio Puerco gage, with Cuba marked.

Figure 1. *[Caption to be written.]*

2 Study Area

2.1 Sandoval County

[Describe the county, its hydrogeologic settings, and its main communities and water sources (Fig. 1).]

2.2 Cuba Focus Area

Cuba lies on the eastern edge of the San Juan Basin. The case study focuses on the Ojo Alamo Sandstone and San Jose Formation near Cuba (U.S. Geological Survey, 2020; Kernodle, 1996). *[Describe Cuba, its water supply, and the aquifers it draws from, and define the boundaries of the focus area.]*

3 Data

Town-specific data for Cuba are limited, so the study relies on public datasets that can be filtered to Sandoval County and the area around Cuba. Eight datasets were collected (Table 1). For seven of them, a one-page brief records what the dataset contains, which research questions it can help answer, which plots to make from it, and what it can't show; a plotting guide compares all seven. The raw files are kept unedited in the project repository (Djordjevic et al., 2026).

3.1 Datasets Used

Groundwater levels. Two datasets measure depth to water below land surface, where a larger number means a lower water table. The USGS statistics (U.S. Geological Survey, 2026c) give long monthly records for 9 wells at four sites: Sandia, Lincoln Middle School in Rio Rancho, Bernalillo, and Dome Road. The same download also covers stream gages, canals, and reservoirs, but only the groundwater records are used. The NM Bureau of Geology measurements (New Mexico Bureau of Geology and Mineral Resources, 2026c) cover more wells but fewer years: most were measured in 1997–1998, and only 5 wells were measured again in 2016–2019, so few of them can show a long-term trend.

Well and aquifer records. The State Engineer's aquifer-test records (New Mexico Office of the State Engineer, 2026b) are the only dataset with hydraulic properties, such as transmissivity and storativity, and they name the formation for 125 of the 179 tests. They include 8 Ojo Alamo Sandstone records at 5 locations, among them tests from 1957 and 1960 at a school well about half a mile from Cuba, but only one San Jose Formation well. The State Engineer's well logs (New Mexico Office of the State Engineer, 2026c) add borehole depth and the depth at which water was first found on the day each well was drilled, which is not a current water level. *[Add the exact download page for the aquifer-test records.]*

Geology of the Cuba-area aquifers. Two publications describe the formations studied for RQ4: a USGS map atlas of the Ojo Alamo Sandstone across the San Juan Basin (Thorn et al., 1990), and a study of the San Jose Formation near Lindrith that identifies four water-bearing units and maps the Cuba Mesa Member (Blandford et al., 2025). Both are maps and reports rather than data tables, and neither shows which formation Cuba's municipal wells draw from.

Surface water at Cuba. USGS station 08332525 on the Rio Puerco (U.S. Geological Survey, 2026b) is the only dataset measured at Cuba itself. Six of its 10 streamflow visits found no flow, and the other four measured 6.9 to 20.4 cubic feet per second. Because these are occasional visits rather than a continuous record, the data are used only as context for the Cuba case study.

Not used. The aquifer sensitivity file (New Mexico Water Data, 2026) has no coordinates or shapes, so it can't be mapped or matched to Cuba without its separate map layer, which was not obtained.

3.2 Data Not Available

The most important missing data are Cuba's municipal well records: well IDs, the formation each well draws from, pumping volumes, and water-level history, held

Table 1. Datasets collected for the study. Distances are measured from Cuba.

Dataset	Source	Contents	Relevance to Cuba	Used for
Groundwater-level statistics	USGS	Monthly and yearly depth to water at 9 wells on 4 sites, 1998–2026	Wells are 40–59 miles away, in the eastern part of the county	RQ1, RQ5
Groundwater levels	NM Bureau of Geology	456 depth-to-water measurements at 87 wells, 1997–2019, with well depth	Nearest well is about 24 miles away	RQ1, RQ2, RQ5
Aquifer tests	NM Office of the State Engineer	179 tests in Sandoval County, 1930–2008: well construction, formation, and hydraulic properties	Includes tests at a well about half a mile from Cuba	RQ2–RQ4
Wells with well logs	NM Office of the State Engineer	Drilling records for 6 Sandoval County wells, 1978–2022: borehole depth and depth to first water	Link to Cuba's wells not verified	RQ2, RQ5
Ojo Alamo Sandstone	USGS map atlas, 1990	Maps of depth to the top of the sandstone, its thickness, and wells completed in it	Regional; covers the Cuba area	RQ4
San Jose Formation near Lindrith	NM Geological Society, 2025	Water-bearing units, thickness of the Cuba Mesa Member, and groundwater levels	Centered near Lindrith, about 20 miles north	RQ4
Rio Puerco at Cuba	USGS station 08332525	10 streamflow measurements (2019–2026) and 4 gage-height readings	At Cuba, but measures the river, not groundwater	RQ3, RQ5
Aquifer sensitivity	NM Water Data	Sensitivity class for 613 mapped areas	Can't be located: the file has no coordinates	Not used

by the Village of Cuba or the State Engineer (New Mexico Office of the State Engineer, 2026a). Every dataset brief identified these records as the missing piece for RQ3. Also not obtained were the USGS digital aquifer data for the San Juan Basin (U.S. Geological Survey, 2020), groundwater-quality records such as the Brackish Water Assessment (New Mexico Bureau of Geology and Mineral Resources, 2026b), acequia locations (New Mexico Interstate Stream Commission, 2026), and records from other monitoring networks (U.S. Geological Survey, 2026a; New Mexico Bureau of Geology and Mineral Resources, 2026a). Section 7 describes what their absence means for the findings.

4 Methods

4.1 Data Cleaning

Cleaning makes the data correct and consistent without drawing conclusions; averaging, trends, and narrowing to the Cuba area happen in the analysis. Raw downloads are kept unedited. A MATLAB script, `cleaning/clean_csvs.m`, reads them and writes the cleaned files (Table 2), so anyone can rerun it and get the same result. As it runs, it prints checks such as row counts, duplicate rows, and the list of formation names.

Shared conventions. Every cleaned file starts with the same six columns: site ID, site name, latitude, longitude, date, and source. Each has one row per measurement, sorted by site and date. Numbers are stored without units or commas, dates are written as YYYY-MM-DD, and column names end with their unit, such as `well_depth_ft`. Missing values are left blank rather than filled with placeholders such as 0 or −9999, which a chart would plot as real values.

Table 2. Cleaned files produced so far.

File	Rows	Contents
groundwater_levels_monthly.csv	2,673	Monthly mean depth to water at 9 wells, June 1998 to September 2026
aquifer_tests.csv	179	One row per aquifer test in Sandoval County, 1930 to 2008, with 17 columns

Groundwater levels. From the USGS statistics, the script keeps the depth-to-groundwater rows that are monthly means and renames the columns to the shared names. Wells with an A or B suffix are kept as separate sites, because they share a location but are measured at different depths. The cleaned file was checked against the raw data: it has the same rows, dates, and values, with no duplicate site-and-date rows.

Aquifer tests. The script keeps all 179 Sandoval County tests, including repeat tests on the same well, and the 16 raw columns the analysis needs: well construction, formation, water level, pumping rate, and hydraulic properties. Empty and administrative columns are dropped. It converts numbers stored as text, standardizes dates, and cleans formation names by trimming spaces, converting them to uppercase, and fixing a misspelling. Three decisions were needed:

- One hydraulic conductivity was recorded as a range, 11 to 21 ft/day; the midpoint, 16 ft/day, is used.
- 104 tests list January 1, 1900, as the date of the starting water level, meaning the date is unknown, so that column is left out.
- 30 tests are dated January 1, which may mean only

Figure 2 (to be added): time series of groundwater levels for wells with repeated measurements.

Figure 2. *[Caption to be written.]*

the year is known; they are kept as is.

Many tests are missing values, such as the formation (54 tests) and storativity (165 tests); these cells are left blank.

Open issues. Most of the State Engineer's well IDs in the aquifer-test file are incomplete: 157 of the 179 are partial, so the cleaned file has 73 site IDs for about 149 distinct wells. The analysis therefore tells wells apart by name and location rather than by ID. The Bureau of Geology water levels, the Rio Puerco measurements, and the well logs still need cleaning; the first and last of these also need their UTM coordinates converted to latitude and longitude. *[Update this paragraph once the remaining files are cleaned.]*

4.2 Analysis

The analysis focuses on measurable indicators, such as depth to water, water-level change over time, well depth, aquifer unit, and hydraulic properties from aquifer tests, rather than a full hydrologic model. Where records are sparse, the results say so instead of filling the gaps. *[Analysis lead: describe the method used for each research question.]*

4.3 Classifying Findings

Each finding is labeled as one of the following:

- **Measured:** a trend or pattern the data show directly.
- **Hypothesis:** a possible explanation the data are consistent with but don't prove.
- **Insufficient data:** a question the available records can't answer yet.

4.4 Interactive Dashboard

Results are presented in an interactive Streamlit dashboard built with pandas and Plotly. Code and documentation are maintained on GitHub (Djordjevic et al., 2026). *[Dashboard lead: describe each page of the app and what a visitor can do there.]*

5 Results

5.1 Groundwater Levels

[RQ1 (Fig. 2).]

5.2 Wells and Aquifer Units

[RQ2.]

5.3 Cuba Case Study

[RQ3 and RQ4. Include the Rio Puerco at Cuba stream measurements as surface-water context, not as evidence about groundwater.]

5.4 Monitoring Coverage and Data Gaps

[RQ5.]

6 Discussion

[What the results mean for Sandoval County and Cuba, and which indicators best communicate groundwater stress (RQ6).]

7 Limitations

The project proposal included three questions that this report does not answer, because the data needed for them were not obtained during the project:

- **Groundwater quality and salinity.** No groundwater-quality records were collected. The water-quality measurements in the USGS file all come from streams, mainly the Rio Grande below Cochiti Dam, far from Cuba.
- **Acequias.** No data on acequia locations or service areas were collected.
- **Groundwater and surface-water links.** Comparing the two requires repeated water-level measurements near a stream. The only stream records at Cuba are 10 visits to the Rio Puerco gage, and the nearest well in the water-level datasets is about 24 miles away.

[Add the other data gaps and what the data can't support.]

8 Conclusions

[Summary of findings and recommended next steps.]

Data and Code Availability

Code and data
github.com/pdjordj11/eng_220_aquifer_project
Interactive dashboard
eng220aquiferproject.streamlit.app
Documentation
pdjordj11.github.io/eng_220_aquifer_project

Author Contributions

P. Djordjevic: documentation, final report, presentation, and video.
D. Gonzales: data acquisition: collected seven datasets and wrote a research brief for each, plus a plotting guide comparing them.
J. Wauneka: data cleaning: wrote the MATLAB script that produces the cleaned groundwater-level and aquifer-test files.
M. Watkins: analysis code and visualizations.
M. Yousef: GitHub workflow and the Streamlit app.

Acknowledgments

[To be written.]

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